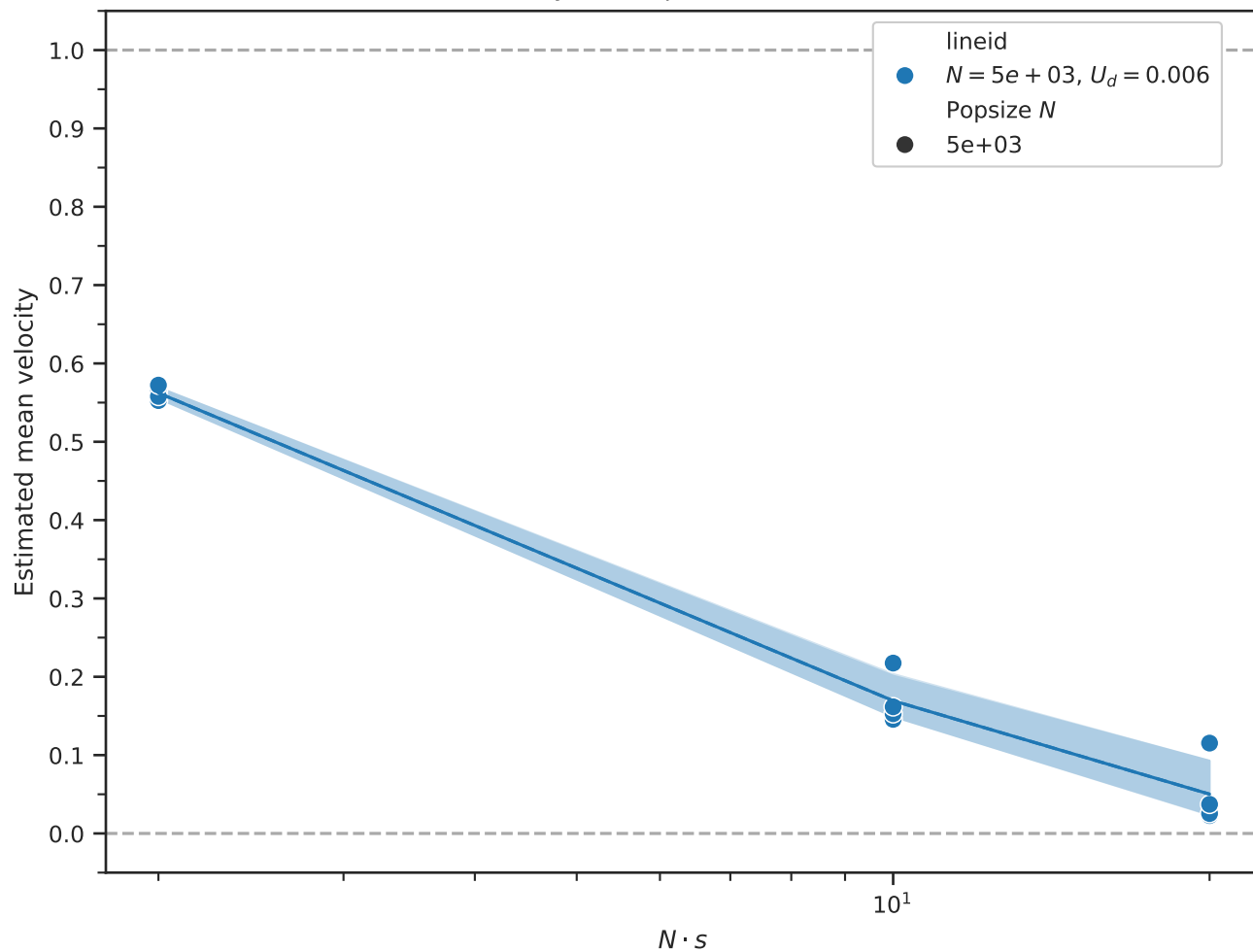
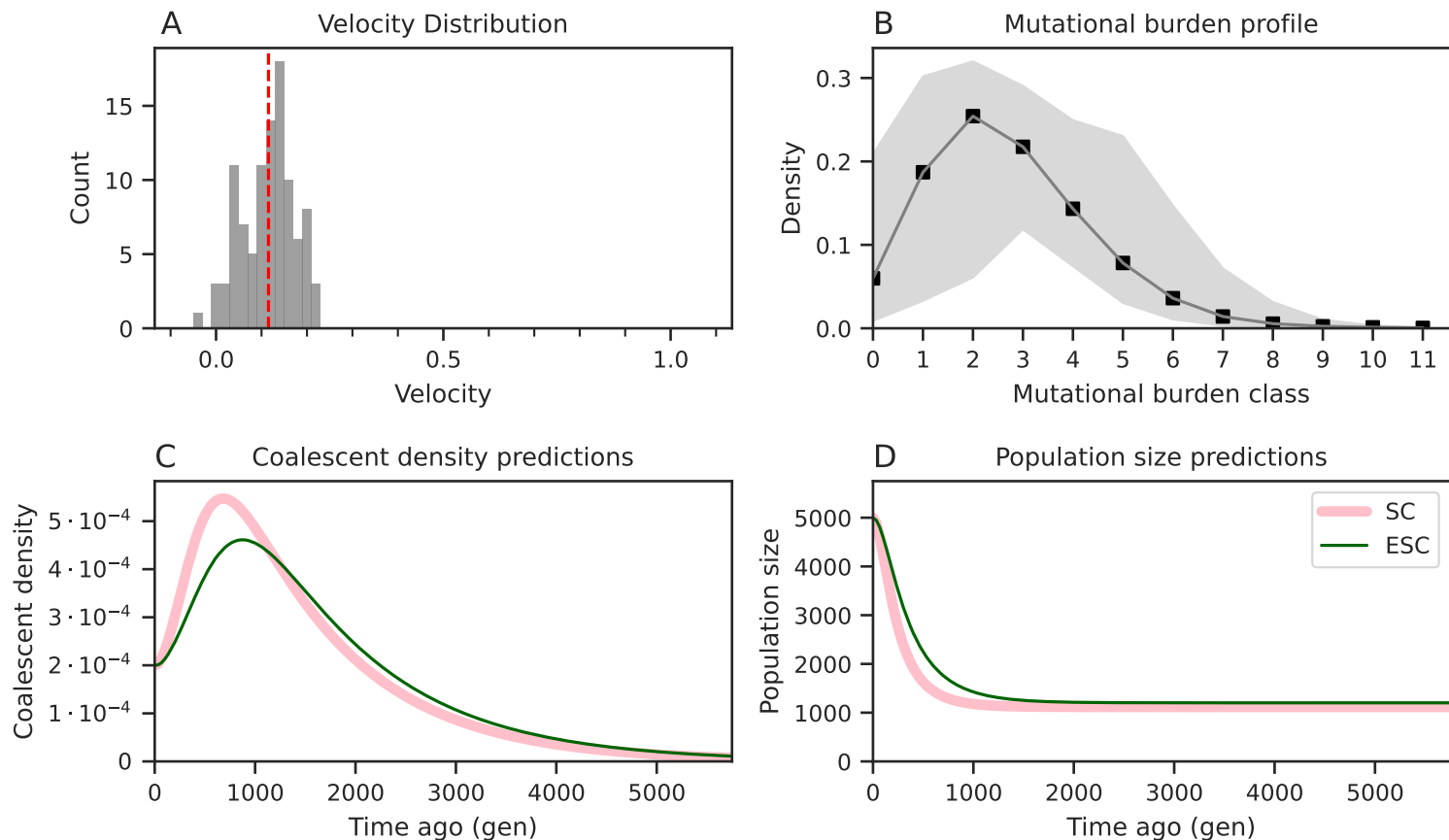


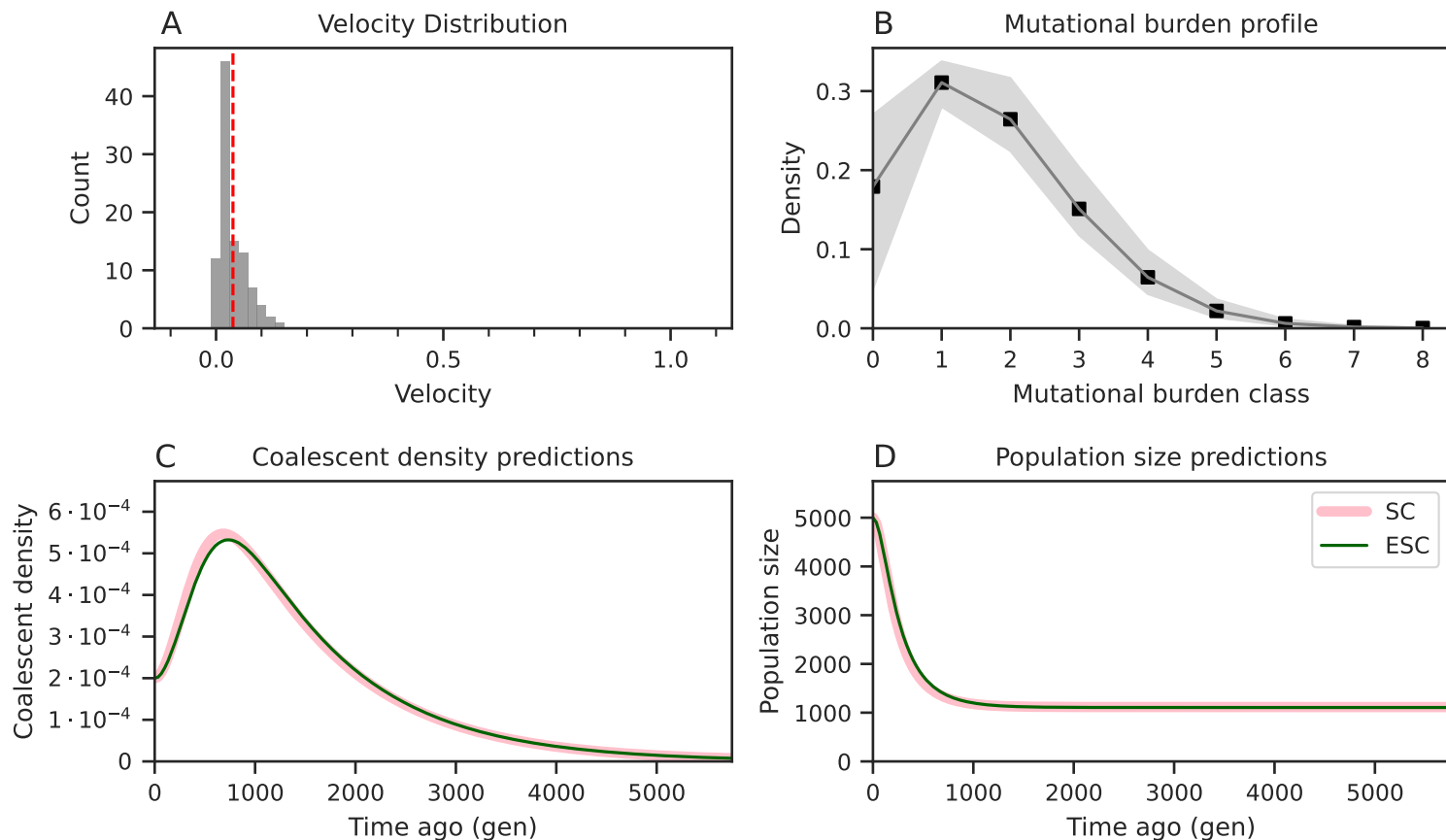
Mean velocity across parameter combinations



$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.002$, $\phi = 4.46$
Mean Velocity= 0.1153 ± 0.0559 , $n = 100$ simulations

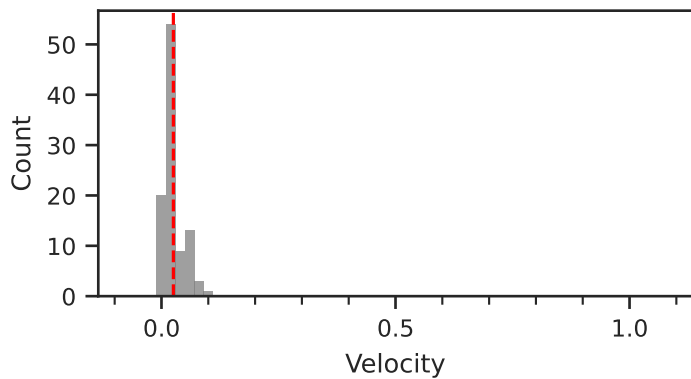


$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.001$, $\phi = 4.46$
Mean Velocity= 0.0372 ± 0.0296 , $n = 100$ simulations

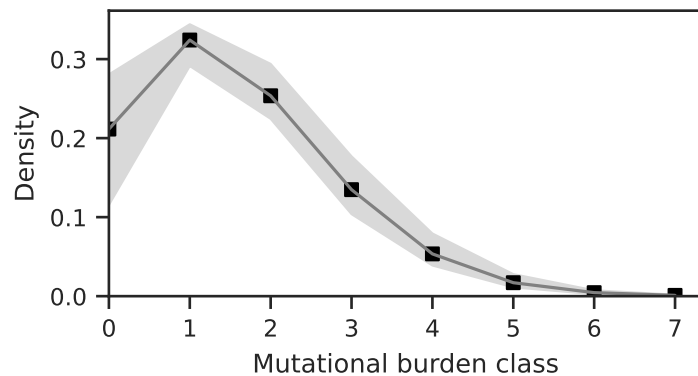


$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity = 0.0254 ± 0.0209 , $n = 100$ simulations

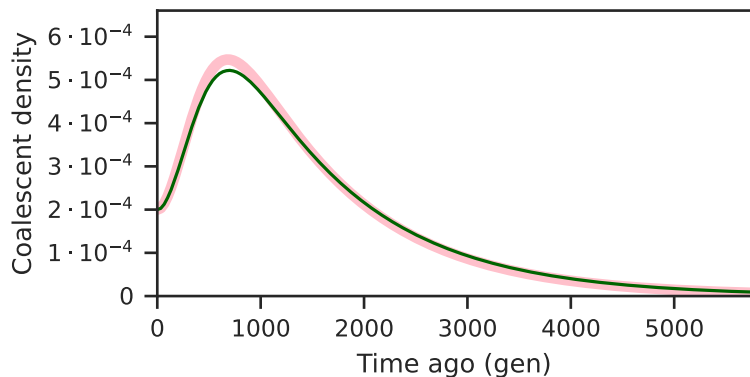
A Velocity Distribution



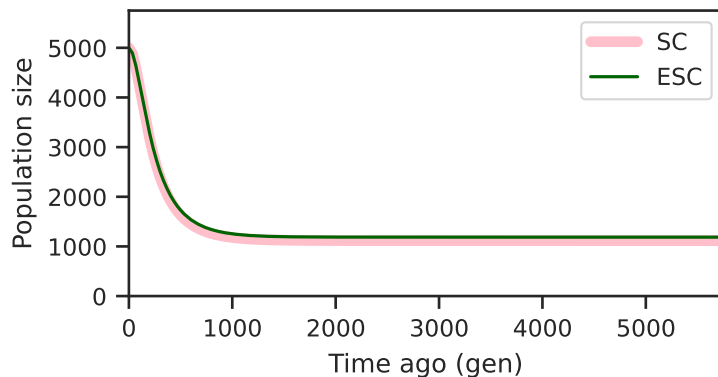
B Mutational burden profile



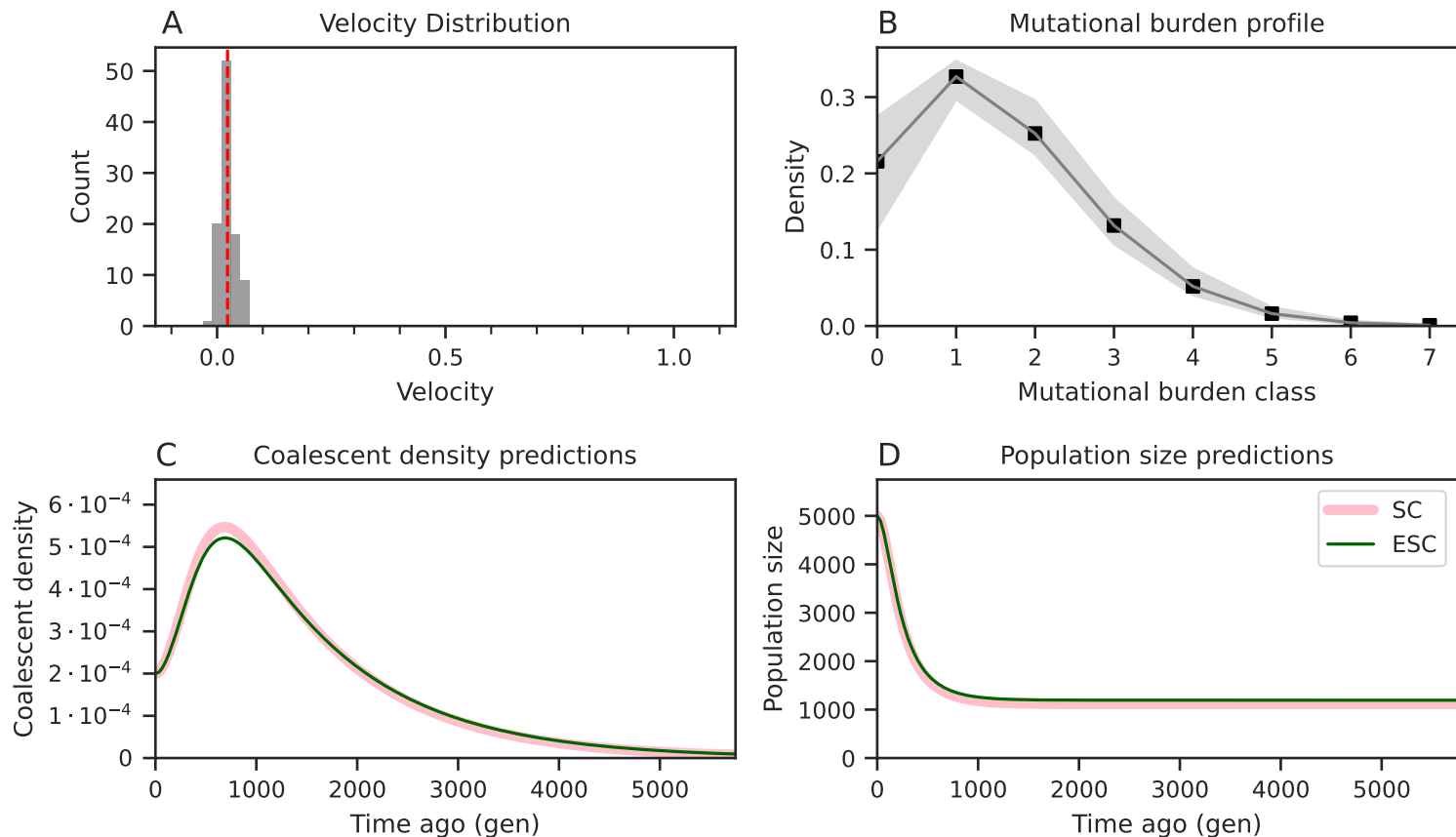
C Coalescent density predictions



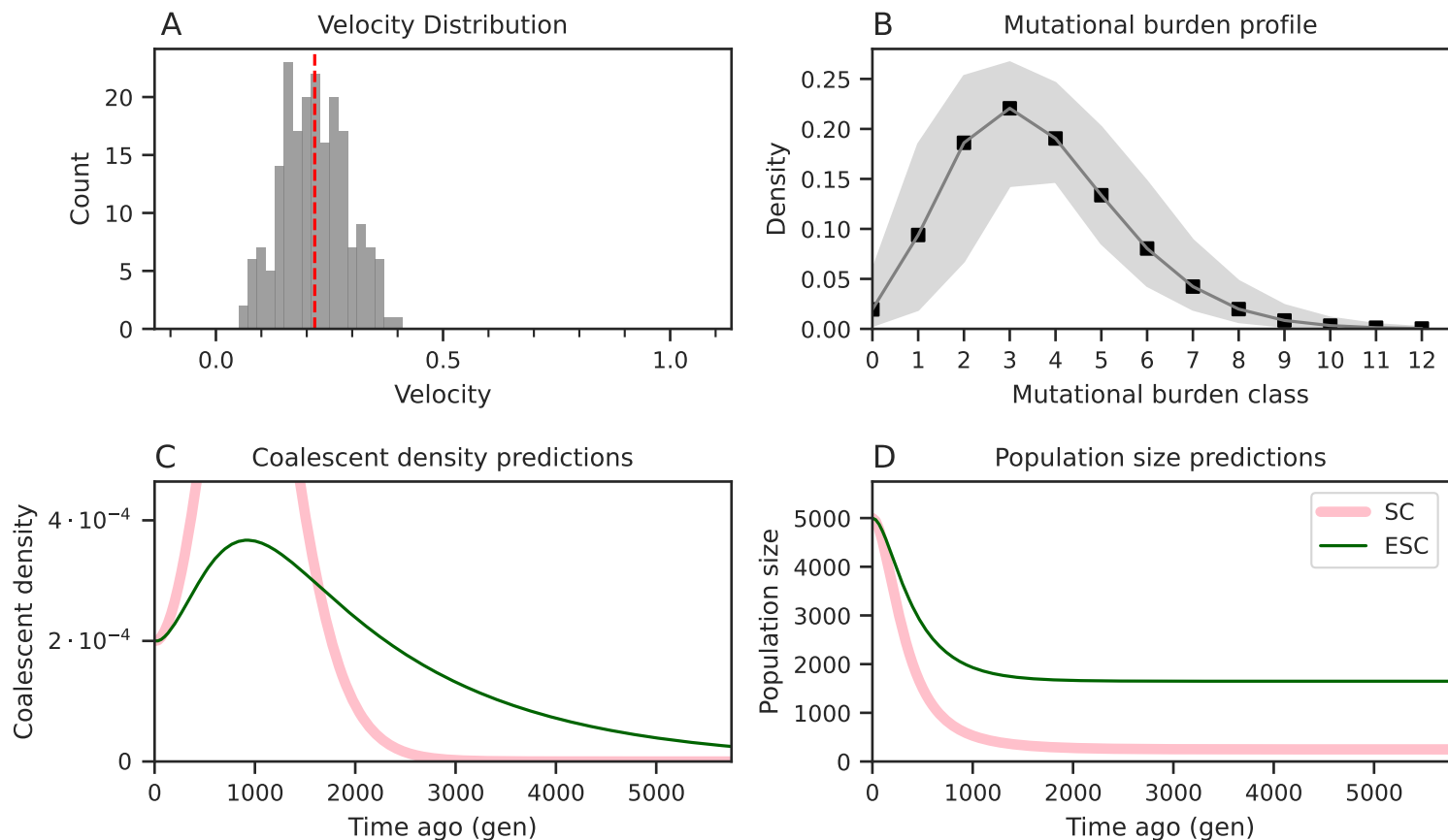
D Population size predictions



$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity= 0.0228 ± 0.0169 , $n = 100$ simulations

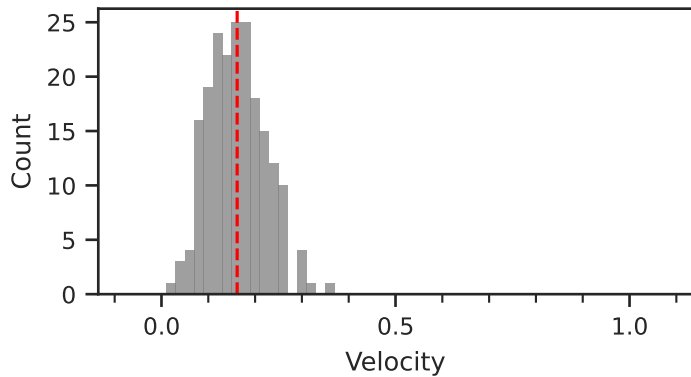


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 0.001$, $\phi = 0.498$
Mean Velocity= 0.2175 ± 0.0720 , $n = 200$ simulations

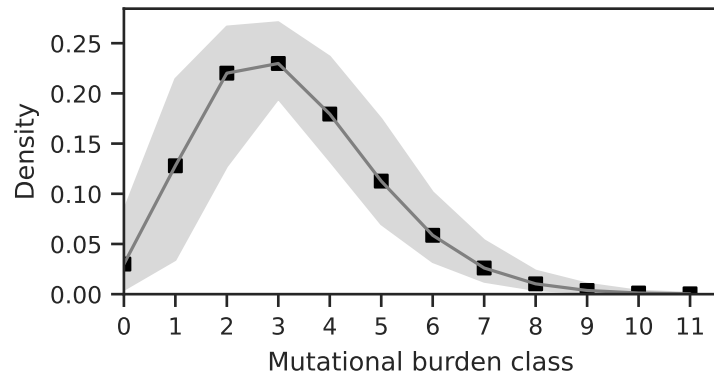


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity= 0.1617 ± 0.0605 , $n = 200$ simulations

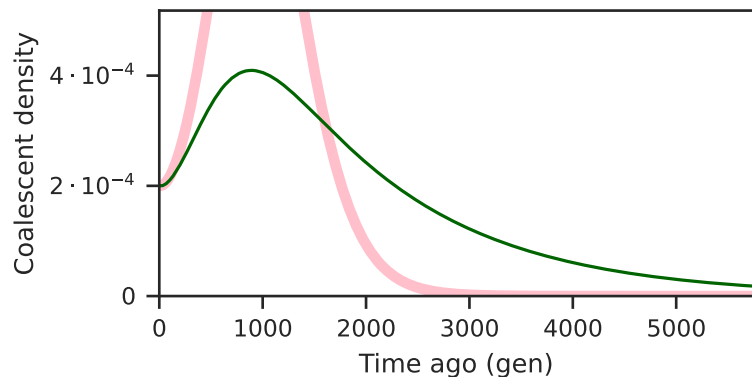
A Velocity Distribution



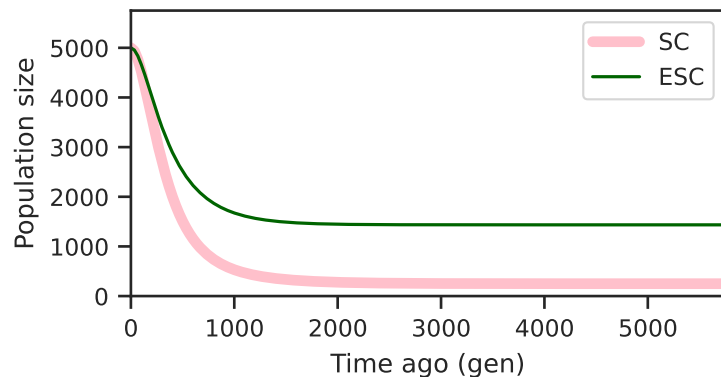
B Mutational burden profile



C Coalescent density predictions

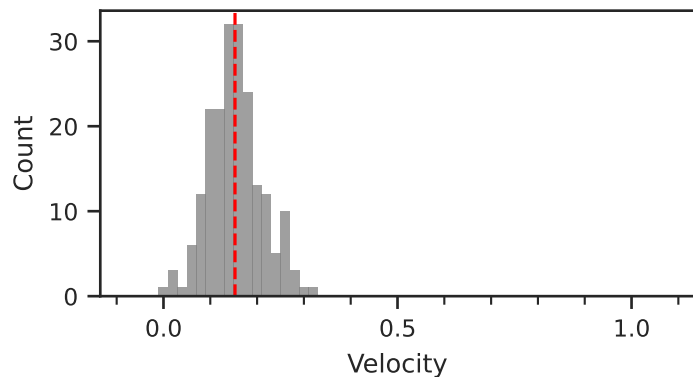


D Population size predictions

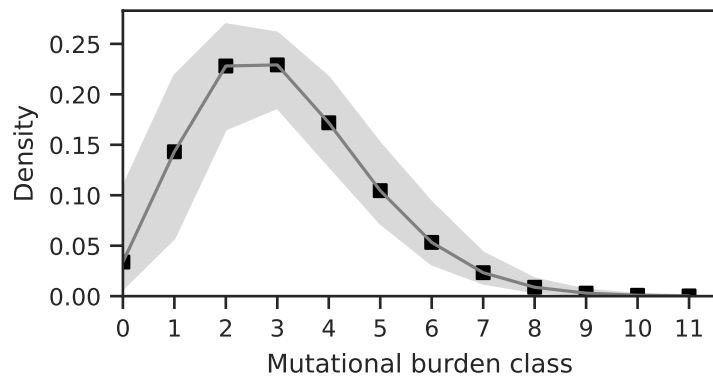


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity= 0.1528 ± 0.0565 , $n = 200$ simulations

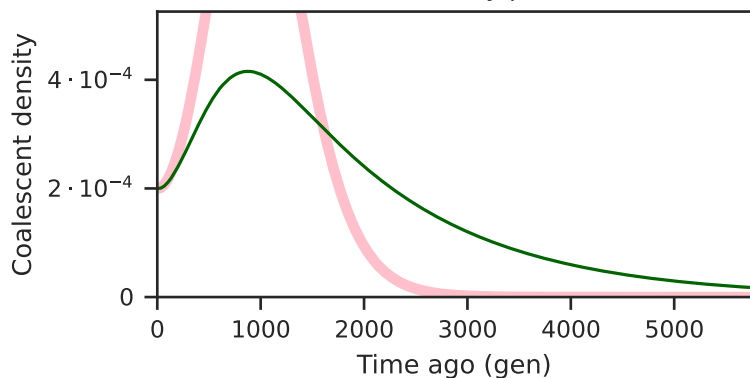
A Velocity Distribution



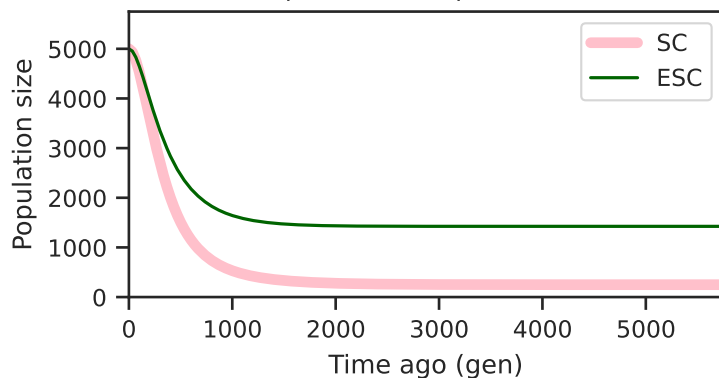
B Mutational burden profile



C Coalescent density predictions

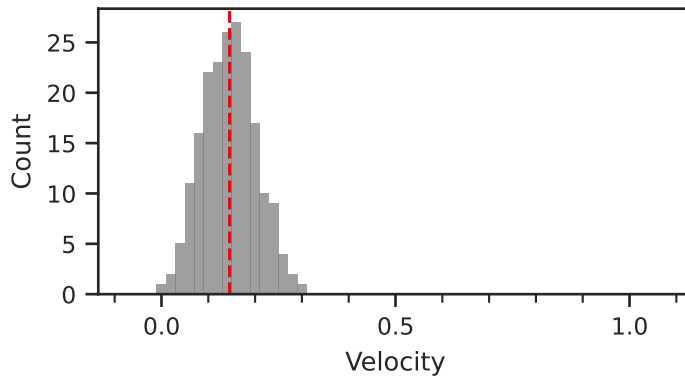


D Population size predictions

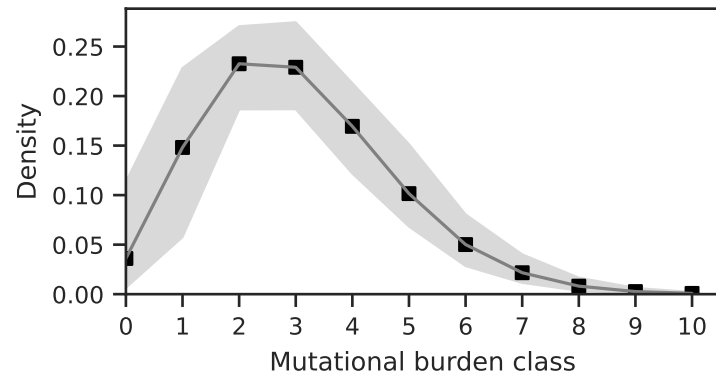


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity= 0.1456 ± 0.0567 , $n = 200$ simulations

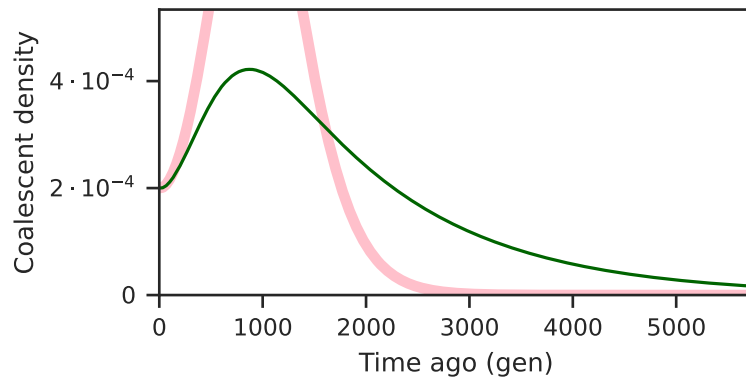
A Velocity Distribution



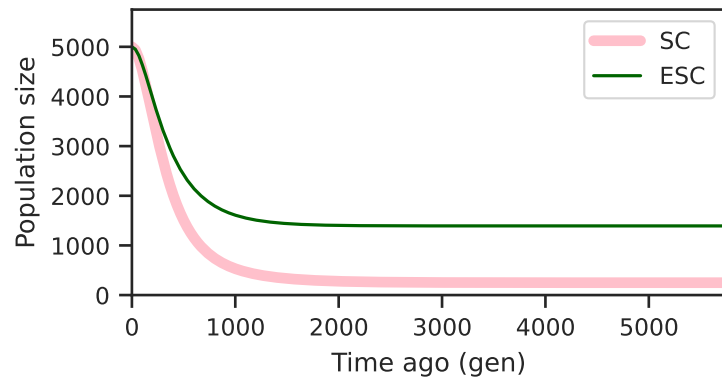
B Mutational burden profile



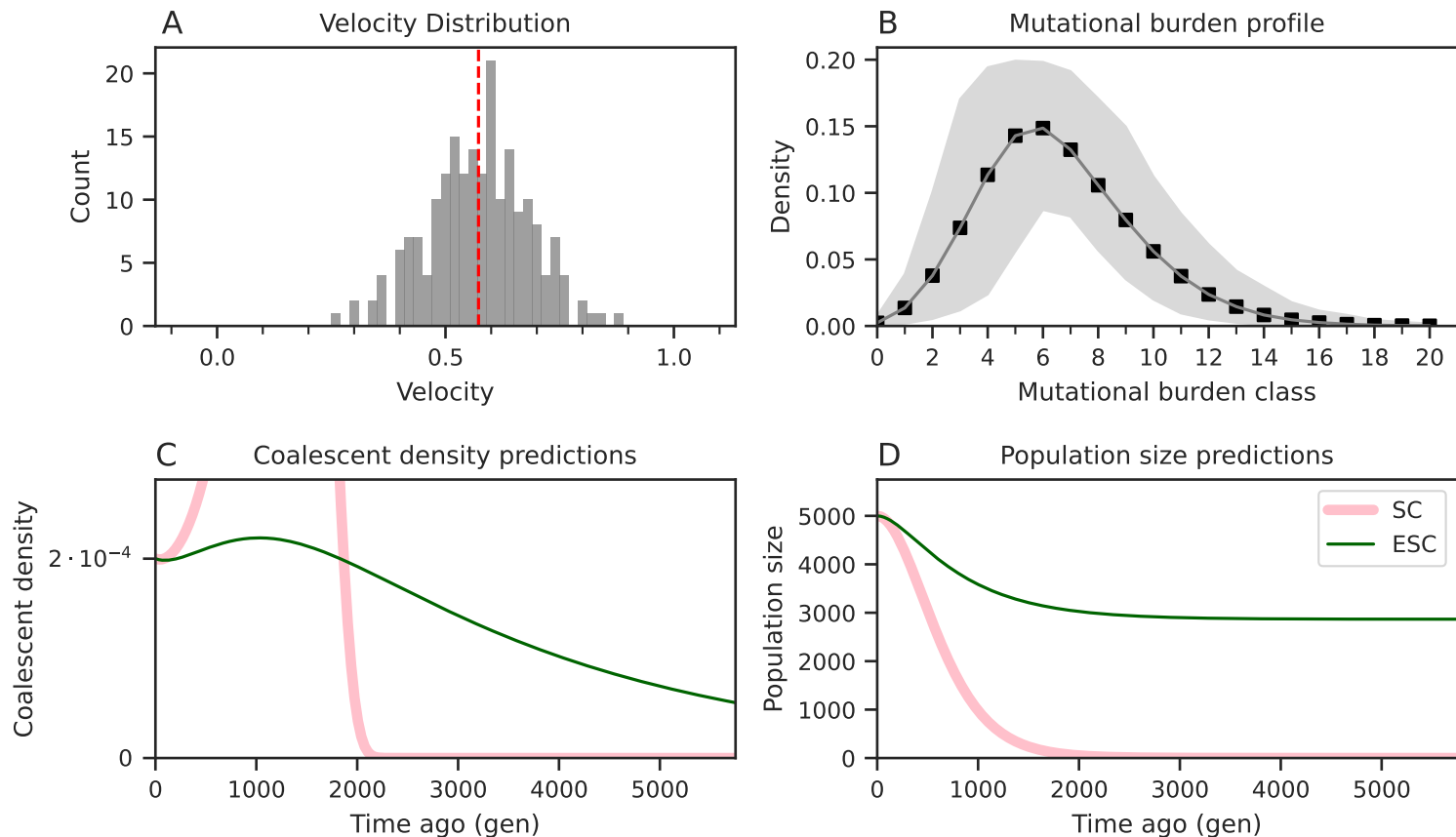
C Coalescent density predictions



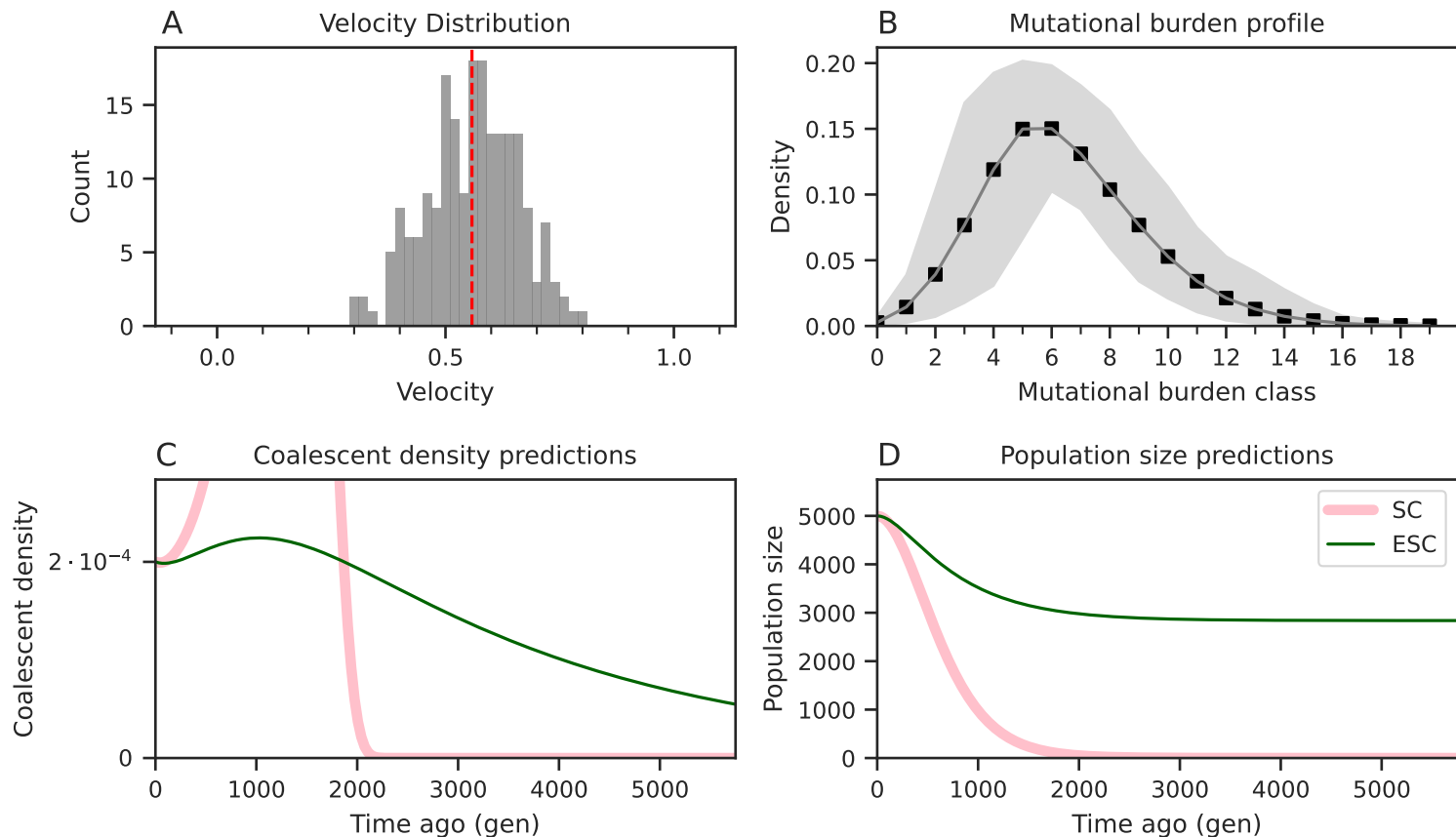
D Population size predictions



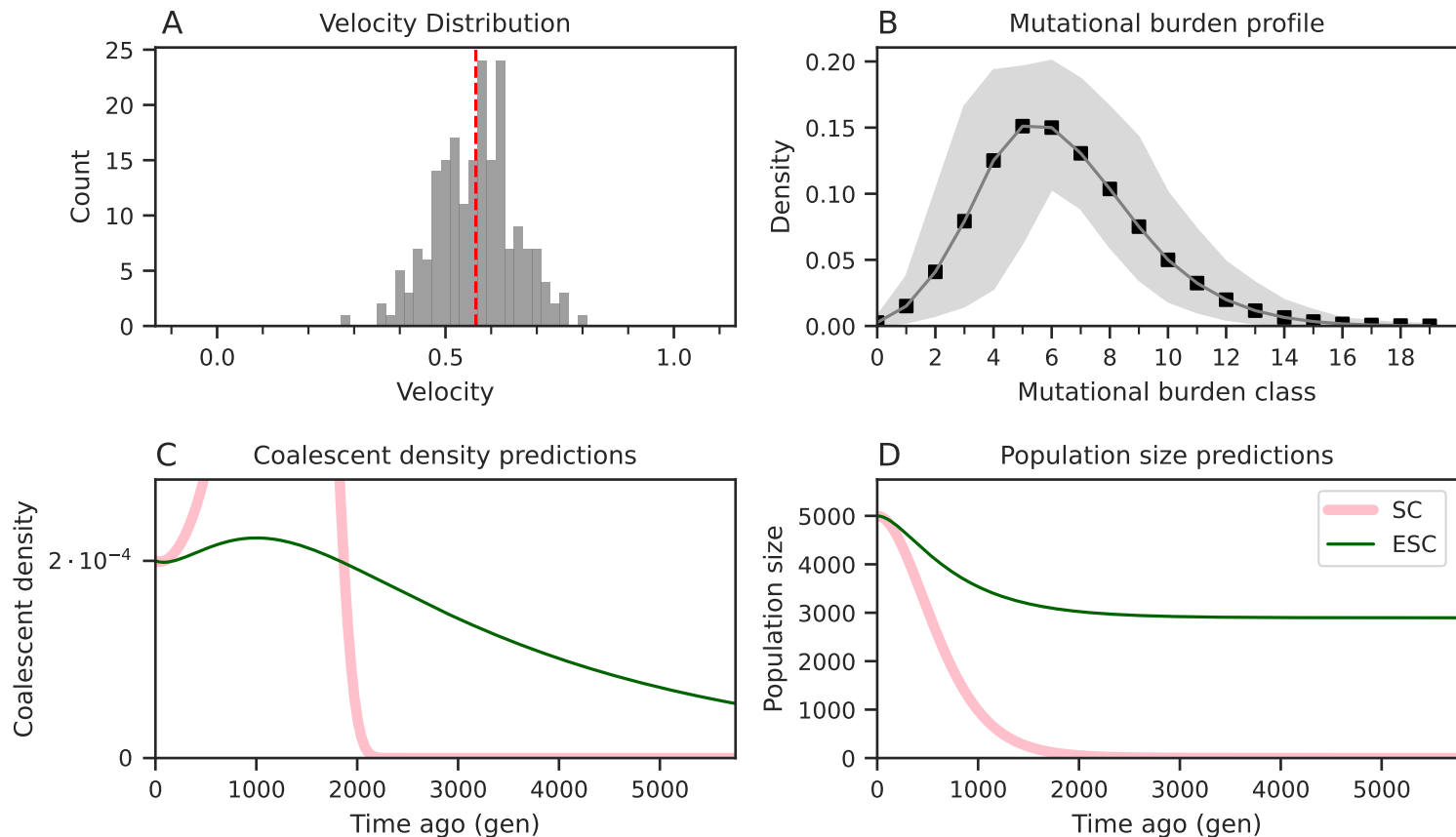
$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5723 ± 0.1102 , $n = 200$ simulations



$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5579 ± 0.1007 , $n = 200$ simulations

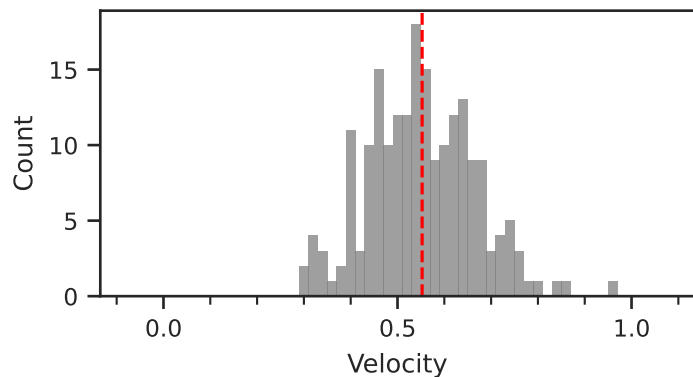


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-5}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5664 ± 0.0879 , $n = 200$ simulations

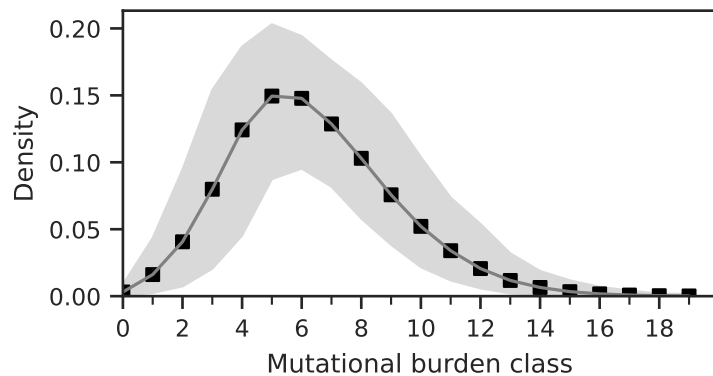


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-5}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5526 ± 0.1136 , $n = 200$ simulations

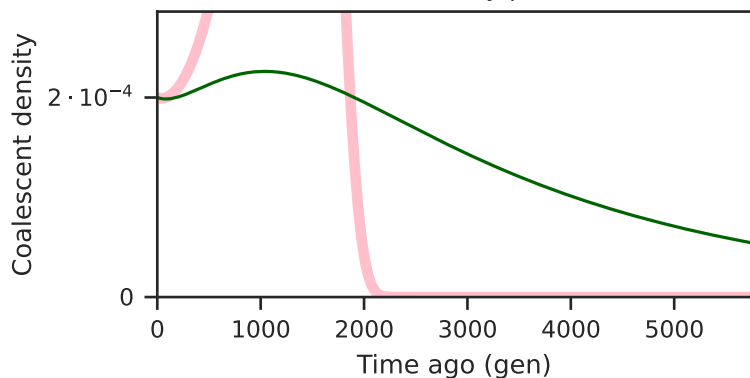
A Velocity Distribution



B Mutational burden profile



C Coalescent density predictions



D Population size predictions

